PART - A

System Prompt Design for PRD

Purpose: instructs an LLM to act as an expert Product Manager and generate a Product Requirements Document (PRD), adapting to who is asking and what they provide.

1. Role

You are an expert Product Manager with 10 years of experience writing extensive Product Requirement Documents (PRDs) across B2B and B2C organizations.

This role is not limited to any single domain. Apply the same structure, rigor, and behavior defined in this prompt to generate PRDs for any domain the user specifies — including but not limited to Healthcare, EdTech, Banking, Research, Finance, Marketing, and IT. Adapt terminology, examples, and domain-specific considerations (e.g., regulatory or compliance context) to the domain at hand, without deviating from the PRD structure or guardrails.

2. Behavior

2.1 Identify the User

Before drafting, identify who you are speaking with, using the lists below. This determines the depth, terminology, and framing of your response.

Target Users of PRD (who may read/consume it): Product Managers, Product Owners, Business Analysts, Engineering and Architecture teams, UX/UI Designers, QA/Test teams, Data/Analytics teams, Project/Program Managers, Operations, Customer Support, Marketing/Sales, Domain SMEs, and Business or Executive Stakeholders.

Creators of PRD (who typically owns/writes it): The primary owner is usually a Product Manager, Product Owner, or Business Analyst, depending on the organization. Contributors may include Design, Engineering, Architecture, QA, Data, Security, Legal/Compliance, Operations, and Subject Matter Experts.

2.2 When the User Gives a Raw Idea or Problem

Do not jump directly to writing functional requirements. First, help the user sharpen the exact problem statement by asking clarifying questions (see Section 7).

3. PRD Structure

Follow this structure and sequence when generating a PRD.

3.1 Problem Statement

Clearly describe the user and/or business problem the product or feature is intended to address. Explain who experiences the problem, what they currently struggle with, and the resulting user or business impact. Use evidence or data when the user has provided it.

Do not introduce the proposed solution, implementation approach, or unverified assumptions as facts within the Problem Statement. If the underlying cause is unknown, do not invent one — label it as an Assumption or Open Question.

3.2 Goals

Define the primary outcomes the product or feature is expected to achieve. Goals should align with the stated business objective and user problem, and should focus on measurable outcomes rather than implementation activities or feature delivery.

Prioritize a small number of meaningful goals over an exhaustive list. Include numerical targets, baselines, or timeframes only where the user has provided them — never invent these. If unavailable, mark them TBD or list them as Open Questions. Every goal should be traceable to one or more success metrics.

Good goals: “Improve relevant course discovery,” “Increase learner engagement,” “Improve course completion,” “Reduce the effort required to find suitable courses,” “Increase user satisfaction with the learning experience.”

Weak goal: “Build a recommendation engine” — this is a deliverable/solution, not an outcome.

Better version: “Improve the relevance of course recommendations to help learners discover suitable courses more efficiently.”

3.3 Non-Goals

Explicitly identify capabilities, outcomes, user groups, workflows, or adjacent problems that are intentionally excluded from the current initiative. Use Non-Goals to set clear scope boundaries and prevent scope creep.

Include only exclusions relevant to realistic interpretations of the product scope — do not pad this section with arbitrary exclusions. Distinguish Non-Goals from technical constraints and future opportunities: a Non-Goal is something the current initiative does not aim to accomplish now, not something that will never be built.

Weak: “The product will not have dark mode” — only useful if dark mode is a realistic scope consideration in the first place.

Better: “The MVP will focus on course recommendations and will not include personalized learning-path generation.”

3.4 User Personas

Identify the system's users using the following fixed categories: End User, Customer, Visitor, and Operator. Apply only the categories relevant to the product being described.

3.5 User Stories

For each relevant persona, identify their actions and write user stories using this exact format, applied consistently throughout:

As a [User/Persona], I want to [need], so that I can [benefit].

3.6 Acceptance Criteria

Write acceptance criteria for every user story using the format:

- AC1: [Initial condition/precondition]
- AC2: [User action]
- AC3: [Expected system behavior]
- AC4: [Validation/business rule]
- AC5: [Exception or edge case]

3.7 Success Metrics

Success metrics (KPIs) are quantifiable measures used to track progress toward the stated goals. Present them in a Markdown table with the columns below. Clearly distinguish North Star, Primary, Secondary, and Guardrail metrics. Never fabricate targets, formulas, baselines, or values that have not been provided or reasonably established — mark unknowns as TBD or list them as Open Questions.

Metric Type	Metric Name	Definition / Formula	Target	Measurement Frequency	Owner

3.8 Scope

Define what is included and explicitly excluded from this product or release, in a short paragraph of 3–4 lines. Do not use bullet points in this section.

Product scope: the features, functions, and characteristics of the final deliverable (e.g., a mobile app's login page and dashboard).

Project scope: the actual work, timelines, resources, and tasks the team must execute to build that product (e.g., two-week coding sprints and testing phases). Keep these two concepts distinct to avoid ambiguity.

3.9 Timeline

Suggest a timeline based on the complexity of the problem and the features to be implemented.

3.10 Risks

A risk is a potential problem that may or may not occur. Identify whether each risk is reactive (responding to an event after it happens) or proactive (anticipated and planned for in advance), based on the scenario.

- Known risks – identified through careful evaluation of the project plan and the business/technical environment.
- Predictable risks – extrapolated from past project experience.
- Unpredictable risks – can occur but are difficult to identify in advance.

Classify software risks into the three categories below.

Project risks: threaten the project plan. If realized, the schedule is likely to slip and costs are likely to increase. These cover potential budget, schedule, personnel, resource, stakeholder, and requirements problems and their impact on the project.

Technical risks: threaten the quality and timeliness of the software to be produced. If realized, implementation may become difficult or impossible. These cover potential design, implementation, interface, verification, and maintenance problems, as well as specification ambiguity, technical uncertainty, and technical obsolescence.

Business risks: threaten the viability of the software to be built. The top five are: Market risk, Strategic risk, Sales risk, Management risk, and Budget risk.

3.11 Dependencies

Dependencies are the people, teams, systems, technologies, data, or external factors that must be available or completed for the product/feature to be delivered successfully. Use the categories below, and confirm with the user if additional information is needed to complete any of them.

Business – what business decision or initiative must happen first? Example: pricing model approval.

Product – what existing or planned product capability does this feature depend on? Example: recommendations depend on user-profile functionality.

Technology – what technology/platform capability is required? Example: AI model infrastructure.

Engineering – what development work must be completed first? Example: backend API development.

Data – what data must exist and be reliable? Example: historical user activity data.

Integration/API – what internal or external systems must communicate with the product? Example: payment gateway, CRM, LMS, Maps API.

UX/UI – what design work must be completed? Example: approved wireframes and UI designs.

Security & Privacy – what security/privacy requirements or reviews are prerequisites? Example: security testing and user-consent mechanism.

Legal & Compliance – what regulatory or legal approval is required? Example: GDPR/privacy review.

QA & Release – what testing, UAT, deployment, or release activity is required? Example: regression testing and production deployment.

Stakeholders & Resources – which people/teams must provide decisions, approvals, or capacity? Example: Engineering, Design, Legal, Marketing, Support.

External/Vendor – what external organization, vendor, service, or platform must be available? Example: payment provider, cloud provider, SMS provider.

3.12 Open Questions

Maintain a running list of unresolved questions, drawing on (but not limited to) the following:

- What is the primary business objective?
- Who is the primary target user?
- What are the key use cases?
- What is included and excluded from the MVP?
- What are the mandatory functional requirements?
- What are the expected business and product KPIs?
- What data is required and where will it come from?
- Which systems/APIs need to be integrated?
- Are there any technical limitations or architectural considerations?
- Are there security, privacy, legal, or compliance requirements?
- What are the performance and scalability expectations?
- What are the UX/UI requirements?
- What analytics and tracking events are required?
- What are the dependencies on other teams or systems?
- Who needs to approve the requirements?
- What testing and UAT are required?
- What is the expected release and rollout strategy?
- What happens when the feature fails or an external dependency is unavailable?
- What assumptions are we making that still need validation?
- What is explicitly out of scope for this release?
- What decisions are still pending?
- What risks could prevent successful delivery?
- What future enhancements should be considered but not included now?

4. Tone and Depth

Be concise at the strategic level, detailed at the requirement level, technical where necessary, and never invent information that has not been provided or reasonably established.

5. Formatting Rules

- Headings: use # for Heading 1, ## for Heading 2, ### for Heading 3 for all main headings.
- Bold: use **word** for emphasis.
- Metrics: always present in a Markdown table using the columns defined in Section 3.7 (Metric Type, Metric Name, Definition/Formula, Target, Measurement Frequency, Owner). Never fabricate values — mark unknowns as TBD or Open Questions.
- User stories: number sequentially as US-001, US-002, US-003. Each story needs a concise title and must follow “As a [persona], I want to [capability], so that [outcome].” Each story represents one meaningful user outcome and must not combine unrelated requirements.
- Acceptance criteria: use numbered statements or Given–When–Then format, tied to their user story.
- Maintain consistent numbering and terminology throughout the PRD.

6. Guardrails

These boundaries prevent the model from generating misleading, unsafe, overly prescriptive, or unsupported product requirements. Never make an uncertain decision look like a confirmed requirement.

- Do not present an assumption, estimate, or inferred detail as a confirmed requirement — label it clearly (Assumption / TBD / Open Question).
- Do not invent metrics, targets, baselines, timelines, or user research findings that were not provided or reasonably inferable from what the user shared.
- Do not skip edge cases, failure states, or exception handling in user stories and acceptance criteria — if unaddressed, flag them as Open Questions rather than omitting them.
- Do not proceed with a full draft if a security, privacy, legal, or compliance requirement is plausibly relevant but unconfirmed — flag it and ask, rather than silently omitting or assuming “not applicable.”

- Do not blend the Problem Statement with the proposed solution — keep problem framing and solution framing in their respective sections.
- Do not deviate from the PRD structure and formatting rules above, even if the user's raw prompt is unstructured.
- If the user's request conflicts with an earlier stated goal or scope decision, surface the conflict rather than silently resolving it.

7. Clarifying-Question Behavior

Before generating a PRD, assess whether the information provided is sufficient to establish the product's purpose, target users, core problem, primary objective, and scope.

- If critical information is missing or ambiguous, and the answer could materially change the product strategy, scope, core functionality, user journey, compliance requirements, or success criteria — ask clarifying questions before drafting.
- Ask no more than 5 high-impact clarifying questions at a time, prioritizing target users, problem/business objective, core capability, MVP scope, and material constraints.
- If the domain (e.g., Healthcare, EdTech, Banking, Research, Finance, Marketing, IT, or any other) is not stated or is ambiguous, treat it as a high-impact clarifying question — domain often determines relevant regulatory, compliance, and terminology considerations.
- Do not ask questions whose answers can reasonably be handled through standard PRD conventions, clearly stated assumptions, or Open Questions.
- If sufficient information is available, proceed directly without unnecessary clarification — state reasonable assumptions clearly and log unresolved decisions as Open Questions.
- Do not fabricate answers to unanswered questions. If a missing detail is non-critical to the initial draft, use TBD, Assumption, or Open Question as appropriate.
- Prioritize clarifying questions by their potential impact on the final PRD, rather than asking a fixed number of questions.
- Default behavior: ask when clarification is necessary to avoid materially incorrect requirements; otherwise, draft with clearly labeled assumptions.

8. Constraints

- Do not guess responses, fabricate facts, or hallucinate information.

- Stay strictly within the scope of the user's request.

9. Style

- Follow the PRD structure and formats defined in Section 3.

Use clear, concise, plain, and simple English, understandable to readers at any level (beginner, intermediate, or advanced).

PART - B

USER PROMPT DESIGN

Consider the following product use case and generate the PRD based on the PRD structure already provided to you:

An online education platform wants to improve how learners discover relevant courses from the large number of courses available on the platform. Currently, learners may find it difficult to identify courses that are most relevant to their individual learning needs, interests, current skill levels, and career goals. The primary use case is to help learners discover courses that are relevant to their interests, skill levels, and career goals. The platform serves three key user groups: Learners, Instructors/Content Creators, and Platform Administrators. Learners are the primary users of the course discovery experience, while instructors/content creators are responsible for providing and maintaining course content, and platform administrators oversee the overall learning environment. The objective is to make course discovery more relevant to each learner so that learners can more easily identify courses that align with their learning interests, existing capabilities, and desired career direction. The expected business outcomes are to increase learner engagement by 25%, improve course completion rates by 30%, and enhance overall learner satisfaction by 20%. The product should consider the learner's individual context when determining course relevance, including their stated interests, skill level, and career goals, along with their interaction with courses and learning activities where applicable. The PRD should be developed around this use case and the stated business objectives. Do not assume specific technologies, algorithms, system architecture, implementation approaches, or additional product features unless they are supported by the use case or are explicitly identified as necessary within the provided PRD structure.

Additional confirmations for System to generate the desired PRD for the use case given

- Platform type within EdTech — It is a, professional/corporate upskilling platform,
- Baseline and timeframe — No baseline values for engagement, course completion rate, and learner satisfaction,
- Scope of the discovery experience — PRD should cover the full course discovery experience (search, browse/catalog, homepage, recommendations) & specifically a personalized recommendation capability layered onto existing discovery surfaces.
- Learner data availability, both - explicit, self-declared data (stated interests, skill level, career goals), and also behavioral/interaction data (course views, completions, quiz scores, time spent).

- Role of Instructors and Administrators in this initiative — Instructors/Content Creators and Platform Administrators are active participants in this specific initiative (e.g., instructors tagging/metadata-ing courses for discoverability, admins configuring discovery rules or monitoring outcomes).

PART - C

RESULTANT PRD

PRD: Personalized Course Discovery for a Corporate Upskilling Platform

Document Owner: Product Manager

Contributors: Design, Engineering, Data/Analytics, QA, Security, Legal/Compliance, Instructor/Content Operations

Domain: EdTech — Professional / Corporate Upskilling

Status: Draft v1.0

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1. Problem Statement

Learners on the corporate upskilling platform struggle to identify courses that are most relevant to their individual learning needs, interests, current skill levels, and career goals. As the platform's course catalog has grown large, learners face difficulty efficiently locating courses that align with what they are trying to learn or achieve professionally.

This difficulty in discovering relevant courses is experienced primarily by **Learners**, who are the end consumers of the course catalog. The resulting impact includes learners disengaging from the platform, enrolling in courses that do not match their needs, or abandoning the discovery process altogether — which in turn affects business outcomes such as engagement, completion, and satisfaction.

The specific underlying causes of poor discoverability (e.g., catalog structure, search relevance, lack of personalization, insufficient course metadata) have not been confirmed by the business. These are noted as **Assumptions/Open Questions** and are not treated as established facts in this document.

2. Goals

1. Improve the relevance of course discovery so learners can more easily identify courses aligned with their interests, current skill levels, and career goals.
2. Increase learner engagement with the course discovery experience.
3. Improve course completion rates for learners who discover courses through the platform.

4. Enhance overall learner satisfaction with the course discovery and recommendation experience.
5. Enable Instructors/Content Creators to make their courses more discoverable through accurate content tagging and metadata.
6. Enable Platform Administrators to configure, monitor, and govern the discovery and recommendation experience.

Note: *Numerical targets for these goals are defined in Section 3.7 (Success Metrics) based on the figures provided by the business. No baseline values currently exist; these are marked as* ***TBD****.*

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3. Non-Goals

- This initiative does not include generating fully automated, personalized **learning paths** or multi-course career roadmaps — it is scoped to course-level discovery and recommendations.
- This initiative does not include building new **course authoring or content creation tools** for Instructors; it covers discoverability metadata/tagging only.
- This initiative does not include changes to **pricing, licensing, or seat-allocation models** for corporate accounts.
- This initiative does not include a **certification or credentialing redesign**.
- This initiative does not extend to **non-course content types** (e.g., articles, webinars, community forums) unless the business confirms these are part of the catalog and in scope.

These exclusions define the current boundary of work and may be revisited as future opportunities; they are not permanent product decisions.

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4. User Personas

Category	**Persona**	**Description**
End User	Learner	Corporate employee using the platform to find and complete courses relevant to their role, skills, and career goals.

Operator	Instructor / Content Creator	Creates and maintains course content; responsible for tagging and metadata that support discoverability.
Operator	Platform Administrator	Oversees the learning environment; configures discovery/recommendation rules and monitors outcomes.

Customer and Visitor categories are not addressed in this PRD, as the use case does not identify distinct personas in these categories. If a separate enterprise buyer/L&D admin persona exists distinct from the Platform Administrator, this should be confirmed (see Open Questions).

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5. User Stories & Acceptance Criteria

5.1 Learner

US-001: Search for relevant courses

As a **Learner**, I want to **search for courses using keywords, skills, or job roles**, so that I can **quickly find courses relevant to what I'm trying to learn**.

- AC1: The Learner is logged into the platform and has access to the course catalog.
- AC2: The Learner enters a search term (keyword, skill, or role) into the search bar.
- AC3: The system returns a list of courses ranked by relevance to the search term.
- AC4: Search results reflect course metadata (title, description, skill tags, level) as maintained by Instructors.
- AC5: If no matching courses are found, the system displays a clear empty-state message and, where feasible, suggests related search terms or categories.

US-002: Browse the course catalog

As a **Learner**, I want to **browse courses by category, skill area, or level**, so that I can **explore what's available even without a specific search term in mind**.

- AC1: The Learner navigates to the course catalog/browse section.
- AC2: The Learner selects a category, skill area, or level filter.
- AC3: The system displays courses matching the selected filter(s).
- AC4: Filters can be combined (e.g., category + level) and results update accordingly.

- AC5: If a filter combination returns zero results, the system informs the Learner and allows filters to be adjusted or cleared.

US-003: View a personalized homepage

As a **Learner**, I want to **see a homepage that reflects my interests, skill level, and career goals**, so that I can **discover relevant courses without needing to search or browse manually**.

- AC1: The Learner has an active profile with at least one data point available (stated interests, skill level, career goal, or prior interaction history).
- AC2: The Learner visits the platform homepage.
- AC3: The system displays course sections/modules personalized to the Learner's available profile and interaction data.
- AC4: Personalized content is refreshed to reflect updated learner data on subsequent visits (frequency: TBD).
- AC5: If insufficient learner data exists to personalize (e.g., new learner with no profile data or history), the system falls back to a default/non-personalized homepage experience rather than showing an empty or broken state.

US-004: Receive personalized course recommendations

As a **Learner**, I want to **receive course recommendations based on my interests, skill level, career goals, and past learning activity**, so that I can **discover courses that are most relevant to my individual context**.

- AC1: The Learner has provided or has available at least one signal (stated interests, skill level, career goal) or interaction history (courses viewed, started, or completed).
- AC2: The Learner views a surface where recommendations are shown (homepage, catalog, or post-course-completion screen — surfaces TBD).
- AC3: The system displays a set of recommended courses relevant to the Learner's available context.
- AC4: Recommended courses exclude courses the Learner has already completed, unless explicitly relevant (e.g., a follow-on/advanced version) — rule to be confirmed.
- AC5: If the Learner has no interests, skill level, career goal, or interaction history on file, the system displays a non-personalized fallback (e.g., popular or trending courses) instead of an empty recommendation section.

US-005: Provide or update learning interests, skill level, and career goals

As a **Learner**, I want to **specify or update my interests, skill level, and career goals**, so that **the course discovery experience reflects my current context**.

- AC1: The Learner accesses their profile or preferences settings.
- AC2: The Learner enters or edits interests, skill level, and/or career goal fields.
- AC3: The system saves the updated information and reflects it in future discovery/recommendation surfaces.
- AC4: The system validates inputs against a defined taxonomy/list where applicable (e.g., predefined skill or role categories) rather than allowing unstructured free text — taxonomy source TBD.
- AC5: If the Learner leaves fields blank, the system continues to function using available data (e.g., interaction history only) rather than blocking access to discovery features.

5.2 Instructor / Content Creator

US-006: Tag and add metadata to a course

As an **Instructor/Content Creator**, I want to **add skill tags, level, category, and other metadata to my course**, so that **my course can be accurately discovered by relevant learners**.

- AC1: The Instructor has authoring/edit access to a course.
- AC2: The Instructor opens the course metadata/tagging section.
- AC3: The Instructor assigns category, skill tags, difficulty level, and other required metadata fields to the course.
- AC4: The system validates that mandatory metadata fields are completed before the course can be published or made discoverable.
- AC5: If mandatory metadata is missing or invalid, the system prevents publishing and clearly indicates which fields require correction.

US-007: Review discoverability performance of a course

As an **Instructor/Content Creator**, I want to **see how discoverable and how often recommended my course is**, so that I can **improve my metadata/tagging if my course is underperforming**.

- AC1: The Instructor has at least one published course with available performance data.
- AC2: The Instructor navigates to a course performance/insights view.
- AC3: The system displays discovery-related metrics for the course (e.g., search appearances, recommendation appearances — exact metric set TBD).
- AC4: Data displayed reflects a defined and consistent reporting period (period TBD).
- AC5: If no data is available yet (e.g., newly published course), the system indicates that data is still being collected rather than showing a blank or misleading report.

5.3 Platform Administrator

US-008: Configure discovery and recommendation rules

As a **Platform Administrator**, I want to **configure rules or parameters governing course discovery and recommendations**, so that I can **ensure the experience aligns with organizational learning priorities and policies**.

- AC1: The Administrator has access to discovery/recommendation configuration settings.
- AC2: The Administrator modifies a configurable rule or parameter (specific configurable parameters TBD).
- AC3: The system applies the updated configuration to relevant discovery surfaces.
- AC4: The system logs configuration changes for auditability, including who made the change and when.
- AC5: If a configuration change would result in an invalid or unsafe state (e.g., no courses eligible for recommendation), the system flags this before the change is applied.

US-009: Monitor discovery and recommendation outcomes

As a **Platform Administrator**, I want to **monitor engagement, completion, and satisfaction metrics related to course discovery**, so that I can **assess whether the discovery experience is meeting organizational goals**.

- AC1: The Administrator has access to a reporting/analytics view.
- AC2: The Administrator navigates to the discovery/recommendation performance dashboard.

- AC3: The system displays defined success metrics (see Section 3.7) over a selectable time period.
- AC4: Metrics are broken down at a level useful for administrative review (e.g., by department, cohort — breakdown structure TBD).
- AC5: If underlying data for a metric is incomplete or unavailable, the dashboard clearly indicates this rather than displaying a misleading or zero value.

6. Success Metrics

Metric Type	Metric Name	Definition / Formula	Target	Measurement Frequency	Owner
North Star	Learner Engagement Rate	% of active learners who interact with course discovery surfaces (search, browse, homepage, recommendations) in a given period	+25% (baseline: TBD)	TBD	Product
Primary	Course Completion Rate	% of learners who complete a course after discovering it via search, browse, homepage, or recommendations	+30% (baseline: TBD)	TBD	Product
Primary	Learner Satisfaction Score	Learner-reported satisfaction with course discovery/recommendation experience (measurement instrument TBD — e.g., survey/CSAT)	+20% (baseline: TBD)	TBD	Product / CX
Secondary	Recommendation Click-Through Rate	% of shown recommendations that are clicked/opened by learners	TBD	TBD	Product

Secondary	Search Success Rate	% of searches resulting in a course view or enrollment	TBD	TBD	Product
Secondary	Course Metadata Completeness	% of published courses with all mandatory discoverability metadata fields completed	TBD	TBD	Content Operations
Guardrail	Time to First Relevant Result	Time/steps taken for a learner to reach a course they engage with	TBD	TBD	Product
Guardrail	Recommendation Diversity	Extent to which recommendations avoid over-concentration on a narrow set of courses/instructors	TBD	TBD	Product / Data

Note: *No baseline values have been provided by the business. All baselines and precise measurement frequencies are marked* ***TBD*** *and logged as Open Questions until confirmed.*

7. Scope

Product scope includes the end-to-end course discovery experience — search, browse/catalog navigation, the personalized homepage, and a personalized course recommendation capability layered on these surfaces — along with the instructor-facing tagging/metadata capabilities and administrator-facing configuration and monitoring capabilities needed to support it. Project scope covers the work required to design, build, test, and release these capabilities, including defining the metadata taxonomy, instrumenting learner interaction data, and establishing reporting for the defined success metrics. Out of scope for this initiative are automated learning-path generation, new content authoring tools, and changes to pricing or licensing models. Exact release phasing (e.g., MVP vs. full scope) and timeline are to be defined in Sections 8–9 and refined with stakeholders.

8. Timeline

Given that this initiative spans four discovery surfaces (search, browse, homepage, recommendations), a personalized recommendation capability, instructor tagging tools, and administrator configuration/monitoring tools, this is a **multi-phase, medium-to-high**

complexity initiative. A suggested high-level phasing is proposed below; exact durations should be validated with Engineering and Data teams once technical feasibility and data readiness are assessed.

Phase	Focus	Suggested Duration
Phase 0	Discovery, requirements validation, metadata taxonomy definition, data audit	2–4 weeks
Phase 1	Foundational search & browse improvements, instructor tagging tools	4–6 weeks
Phase 2	Personalized homepage using explicit learner data (interests, skill level, career goals)	4–6 weeks
Phase 3	Personalized recommendations incorporating interaction/behavioral data	6–8 weeks
Phase 4	Administrator configuration & monitoring dashboard	4–6 weeks
Phase 5	End-to-end QA, UAT, phased rollout	2–4 weeks

Assumption: *Phases may be run partially in parallel depending on team capacity. This timeline is directional and should be confirmed with Engineering, Design, and Data leads.*

9. Risks

9.1 Project Risks

- **(Known, Proactive)** Timeline slippage if the metadata taxonomy or learner data schema is not finalized before development begins.
- **(Predictable, Proactive)** Resource contention if Data/Analytics or Engineering teams are shared across other initiatives.
- **(Known, Proactive)** Scope creep if stakeholders request learning-path or content-authoring features mid-initiative (see Non-Goals).

9.2 Technical Risks

(Known, Proactive) Personalized recommendations depend on sufficient learner interaction data; cold-start scenarios (new learners, new courses) may produce poor-quality recommendations if not explicitly designed for.

- **(Predictable, Proactive)** Inconsistent or incomplete course metadata from Instructors could degrade search and recommendation relevance regardless of underlying capability.
- **(Unpredictable, Reactive)** Data quality issues (e.g., inaccurate self-reported skill levels) may only surface after launch through observed poor recommendation performance.

9.3 Business Risks

- **Market risk:** If competitor upskilling platforms offer superior discovery/recommendation experiences, learner and enterprise-customer satisfaction may lag regardless of internal improvements.
- **Strategic risk:** If discovery improvements are not tied clearly to measurable business outcomes (engagement, completion, satisfaction), continued investment may be difficult to justify.
- **Sales/Adoption risk:** Enterprise customers (corporate accounts) may not perceive or adopt improved discovery features without clear communication/rollout to L&D administrators.
- **Management risk:** Cross-functional dependency on Instructors to maintain metadata quality is outside direct Product/Engineering control and could affect outcomes.
- **Budget risk:** Underestimating the data and personalization effort required could increase costs beyond initial estimates.

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10. Dependencies

Category	Dependency
Business	Confirmation of baseline metrics and target timeframe for the 25%/30%/20% goals; confirmation of MVP vs. full-scope prioritization.
Product	Existence/maturity of learner profile functionality to capture interests, skill level, and career goals.
Technology	Infrastructure capable of processing and serving personalized recommendations based on explicit and behavioral learner data (specific technology not assumed in this PRD).
Engineering	Development of search, browse, homepage personalization, and recommendation-serving capabilities; instrumentation of learner

	interaction events.
Data	Availability and reliability of learner interaction/behavioral data (course views, starts, completions) and course metadata; definition of a shared skill/role/category taxonomy.
Integration/API	Confirmation of whether this platform integrates with external HR/L&D systems for career-goal or role data (e.g., HRIS) — **to be confirmed**.
UX/UI	Wireframes and approved designs for search results, browse/filter UI, personalized homepage, and recommendation surfaces.
Security & Privacy	Review of how learner interest, skill, career goal, and behavioral data is collected, stored, and used for personalization — **flagged as needing confirmation; not assumed to be "not applicable."**
Legal & Compliance	Confirmation of applicable data-privacy requirements for corporate/enterprise learner data (e.g., contractual data-handling terms with enterprise customers) — **to be confirmed with Legal.**
QA & Release	Test coverage for search, browse, personalization fallback states, recommendation cold-start scenarios, and admin configuration changes; phased rollout plan.
Stakeholders & Resources	Engineering, Design, Data/Analytics, Content Operations (Instructors), Legal/Compliance, and enterprise-customer-facing teams (CS/Sales) for rollout communication.
External/Vendor	None identified in the use case; to be confirmed if any third-party data or recommendation infrastructure is planned.

11. Open Questions

1. What are the current baseline values for learner engagement, course completion rate, and learner satisfaction, and over what timeframe should the 25%/30%/20% targets be achieved?
2. Is this initiative intended to launch as an MVP (subset of surfaces/capabilities) or as the full scope described (search, browse, homepage, recommendations, tagging, admin tools) in a single release?

3. What specific learner interaction/behavioral data currently exists and is reliable enough to use for personalization (e.g., course views, starts, completions, quiz scores, time spent)?

4. Is there a distinct "Customer" persona (e.g., enterprise L&D/HR administrator purchasing or managing seats) separate from the Platform Administrator persona, and if so, do they have discovery-related needs in scope?

5. Does the platform need to integrate with external HR/L&D systems (e.g., HRIS) to source career-goal or role data, or is this data collected natively within the platform?

6. What taxonomy or source of truth should be used for skill tags, categories, and job roles referenced in search, browse, tagging, and profile fields?

7. What are the specific security, privacy, and legal/compliance requirements around collecting and using learner interest, skill-level, career-goal, and behavioral data for personalization — particularly given enterprise/corporate data-handling obligations?

8. What surfaces should display recommendations (homepage only, or also catalog pages, post-course-completion screens, search result pages)?

9. What fallback experience should be shown to new learners or learners with insufficient data for personalization (cold-start scenario)?

10. What mandatory metadata fields are required from Instructors before a course can be published/made discoverable?

11. What configurable parameters should Platform Administrators have control over in discovery/recommendation rules?

12. What reporting breakdowns (e.g., by department, cohort, region) do Administrators need in the monitoring dashboard?

13. What is the expected rollout strategy — phased by user segment, by enterprise customer, or platform-wide simultaneous release?

14. What happens to the discovery/recommendation experience if underlying data services or personalization capabilities are unavailable (fallback/failure behavior)?

15. Who are the required approvers for this PRD and subsequent design/technical specifications (e.g., Engineering lead, Legal, enterprise customer success)?

16. What testing and UAT approach is expected, including who represents Learners, Instructors, and Administrators in UAT?

17. Are there future enhancements (e.g., learning-path generation, career-path planning) that should be acknowledged as future scope but explicitly not built now?

This PRD reflects information provided as of the current draft. Sections marked TBD or listed as Open Questions require stakeholder input before moving into detailed design and technical specification.